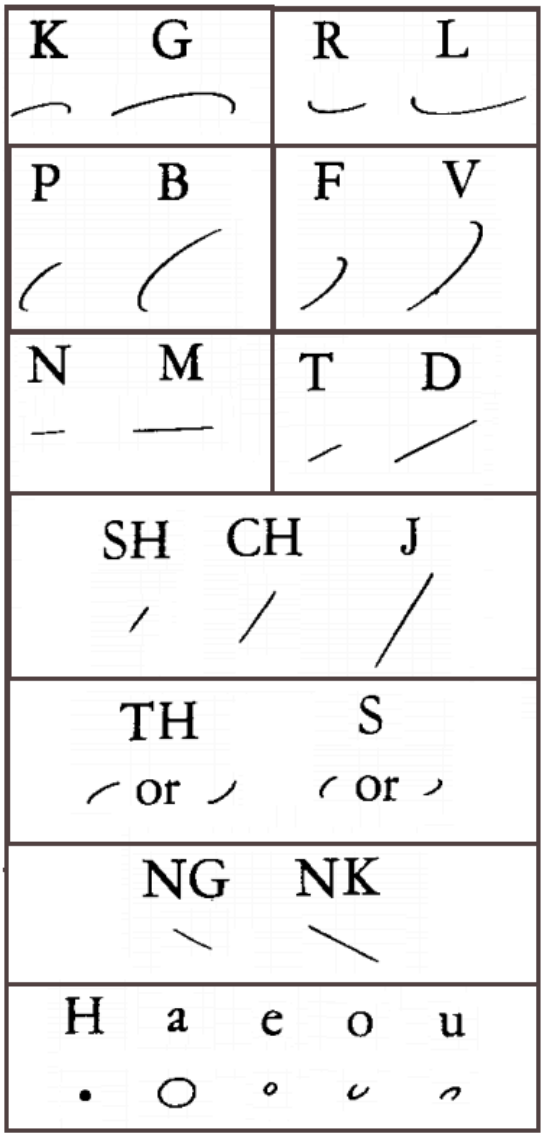
Short Handed

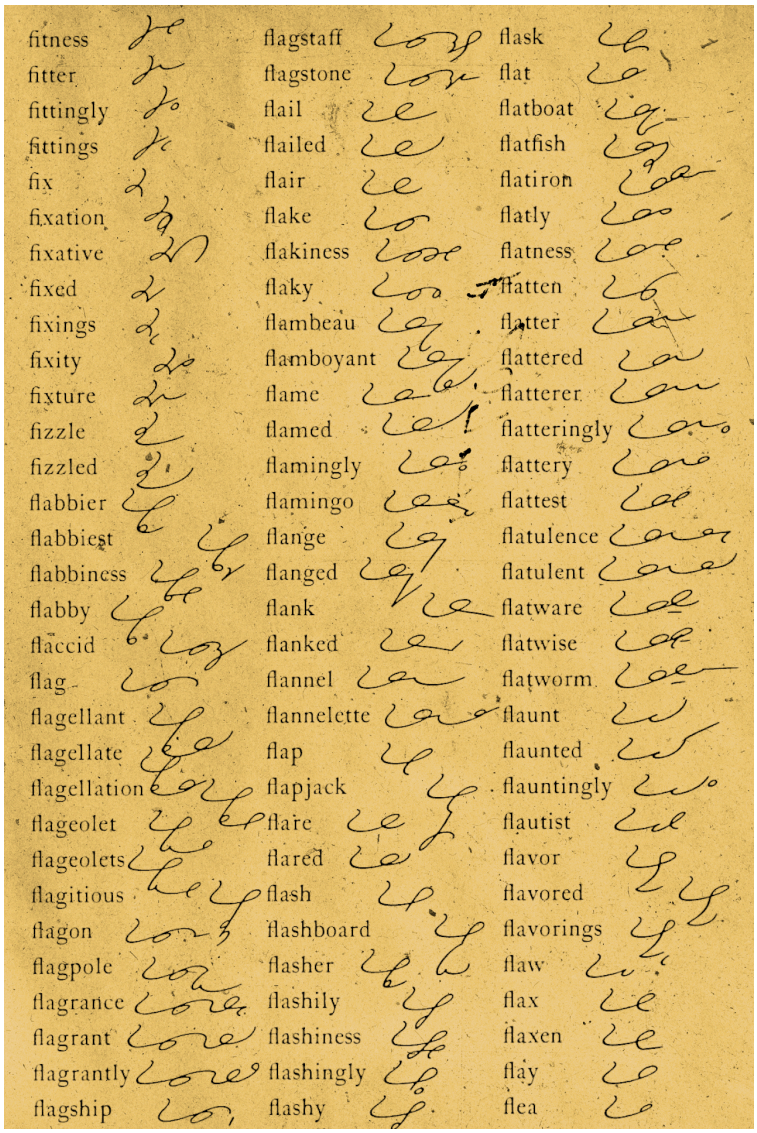
UNG Goldrush Gauntlet 2026 CTF

Solves: 8
Points: 100
Challenge Author: Parker @pforce139
Category: Cryptography
Description: We are short on hands right now, do you think you could decipher this code for us? We scanned this page that we found on site as well as an alphabet that we have determined is accurate. Decipher this back into common day English. format: ggctf{spaces_are_underscores}
Flag: `ggctf{gregg_is_great}`

The encoded message:

The alphabet and dictionary page reference:





The recovered page immediately identified the writing system: a page from a **Gregg shorthand dictionary**, listing words alphabetically (from "fitness" to "flea") with their shorthand outlines. This isn't a custom cipher, it's a real stenography system invented in 1888 for rapid transcription.

Key features from the alphabet:

- Consonant pairs share the same shape but differ in size (K/G, R/L, P/B, F/V, N/M, T/D)
- Vowels are small circles and hooks between consonants (a = large circle, e = small circle, o/u = hooks)
- H is a single dot
- Blends like SH, CH, TH have dedicated strokes
- Common words have "brief forms" where one stroke represents an entire word

Decoding Process

The unknown code contains **5 groups** separated by spaces, with a colon after the second group: **WORD1 WORD2: WORD3 WORD4 WORD5**. I didn't do the words in order, as word 2 was very easy to understand, but the others... took a bit more time.

Word 2: "flag"

The easiest starting point as we can compare with the recovered dictionary page and find a match for "flag." The two dots following it are simply a colon and are not to be confused with an H.

Word 1: "secret"

The first word starts with an S curve (comma shape), followed by an E circle, K stroke, R curve, and T diagonal. Cross-referencing stroke shapes with the alphabet confirmed each component. The word "secret" matched perfectly and made sense.

Word 4: Single stroke

A tiny single stroke sitting between words 3 and 5. In Gregg, many common words are abbreviated to single strokes called "brief forms." This particular stroke matched the brief form for "is."

Word 3: "gregg"

Starts with a G curve (long version of K), followed by R, then E, and two more G strokes. A meta reference to the shorthand system itself.

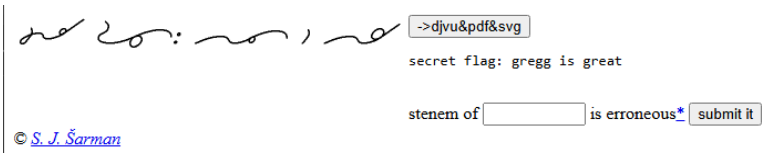
Word 5: "great"

G curve + R curve + E circle + A circle + T diagonal. Confirmed by comparing against similar patterns in the Gregg Shorthand Dictionary PDF.

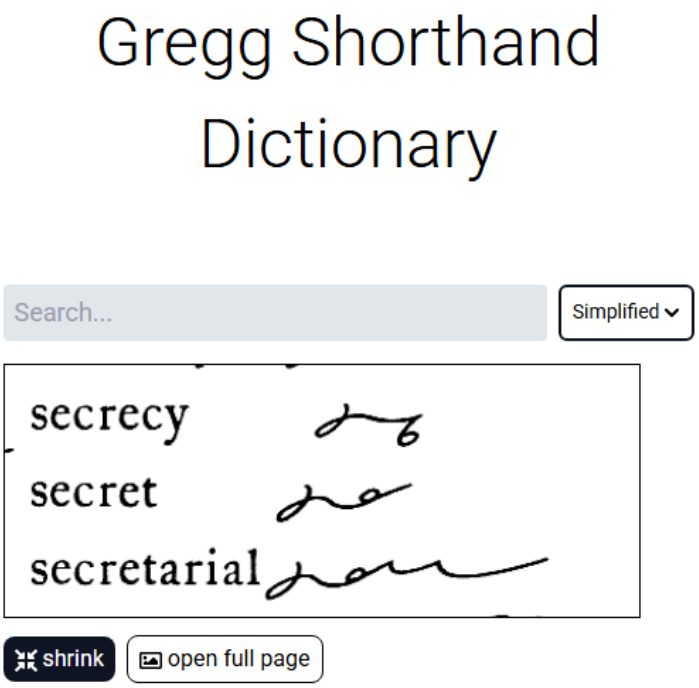
Full message: "secret flag: gregg is great"

Tools and References

- Gregg Shorthand Translator - useful for verifying individual word outlines



- Gregg Shorthand Dictionary Search - searchable dictionary with shorthand outlines



- Gregg Shorthand Dictionary PDF

Flag

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ggctf{gregg_is_great}
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The "secret flag:" portion was not apart of the actual flag and was rather just a nudge to show you that you had solved the challenge.

Lessons Learned

1. Gregg shorthand encodes **sounds** not letters, so thinking phonetically helps with interpretation
2. Brief forms (single strokes for common words like "is", "for", "the") are easy to get and can help with understanding the surrounding content.
3. Start with identifying key parts of the word, like vowels, and then putting it together from there, and remember again that it is based on **sound**!